

Timothy S. Gilbert

1816 E Waverly St, Tucson, AZ 85719 · (520) 395-5640 · tsgilbert@arizona.edu

[Portfolio](#) · [GitHub](#) · [Desert Data Labs](#)

PROFESSIONAL SUMMARY

Ecologist, data engineer and founder of Desert Data Labs LLC, with 8+ years across field ecology, research data management, and full-stack data application development. Specializes in R Shiny applications, SQL database management, and ETL pipelines. Completing an M.S. in Data Science at the University of Arizona.

RELEVANT EXPERIENCE

Founder & Principal Data Engineer

Desert Data Labs LLC: Tucson, AZ — 2026–Present

- Founded Arizona LLC delivering custom dashboards, databases, and data pipelines for small businesses, nonprofits, and research teams without in-house software support
- Design and build production R Shiny applications, SQL databases (Azure SQL, PostgreSQL, SQL Server, SQLite), and ETL/integration pipelines with embedded QA/QC
- Developing EcoPlot Mobile, an offline-capable field-data collection web app (PWA) on Azure (Static Web Apps + managed SQL), as a modern successor to legacy desktop field tools ([click for demo](#))
- Build and maintain the company marketing site in Quarto with automated CI/CD deployment via GitHub Actions
- Established the service catalog, fixed-scope SOW process, hosting/maintenance tiers, and a sliding-scale “Mission Pricing” program providing reduced-rate and pro bono data infrastructure to under-resourced ecology and conservation labs

IT Support Computing Analyst I–III

University of Arizona, VGS Project: Tucson, AZ — March 2022–Present;

promoted from Analyst I to Analyst III Nov 2022, full time

- Developed and deployed R Shiny applications for research data workflows, including data ingestion, management, manipulation, quality assurance, and interactive visualizations
- Designed ETL pipelines for historical ecological datasets, parsing Excel files into structured SQL tables with QA/QC protocols built in
- Maintained local and server-side databases (Microsoft SQL Server, SQLite, Azure Data Studio, MySQL); automated SQL scripts for data quality and reporting
- Conducted 20+ training sessions for USFS and NRCS staff at field offices, demonstrating VGS platform workflows, instructing on data collection best practices, and delivering technical support and field-device maintenance
- Led and mentored a student data entry team, standardizing QA protocols for high-volume historical data transcription
- Supported cross-platform software testing (Windows, iOS, Android) in a C# environment using Visual Studio and VS Code; tracked bugs and fixes via Jira
- Supported agile development workflows and sprint tasks across technical and field teams; led daily scrum meetings
- Assisted in preparing quarterly reports for federal funding sources, documenting project progress, deliverables, and achievements

Rangeland Specialist

University of Arizona, Gila County Cooperative Extension (Rangeland Monitoring): Tucson, AZ — Oct 2017–Apr 2022, seasonal (Sept–Apr), full time

- Conducted long-term vegetation monitoring across rangeland sites using VGS protocols (line intercept, plant diversity, cover, biomass production)
- Provided testing feedback and field validation for VGS platform development, improving usability and data accuracy, eventually leading to a role on the VGS team (above)
- Supported ecological data collection and pasture assessments within the Tonto National Forest in coordination with federal land managers
- Trained USFS employees and local ranchers in standardized plant data collection methods; facilitated collaborative monitoring efforts
- Managed data entry for historical data
- Led seasonal field staff in vegetation monitoring efforts

Lead Aquatics Field Ecologist

National Ecological Observatory Network (NEON): Tucson, AZ — March 2018–Nov 2021, seasonal (Mar–Dec), full time

- Led field sampling campaigns across aquatic and terrestrial systems (water chemistry, invertebrates, flora, and fauna) at Sycamore Creek, Santa Rita Experimental Range, and Jornada Experimental Range
- Collected water-quality metrics, microbial DNA, stream discharge measurements, electrofishing data, and sediment profiles; processed fish vouchers and soil samples for downstream identification and analysis
- Executed terrestrial protocols including small mammal trapping (tagging, biometrics, hair, blood, and ear samples), ground beetle surveys, vegetation structure assessments, and litter trap deployment
- Performed laboratory analysis under sterile conditions, titrations, and reagent preparation; cleaned, sterilized, and maintained lab glassware and equipment
- Contributed to drafting and revising NEON Standard Operating Procedures (SOPs) and followed strict protocol guidelines to ensure data quality
- Managed inventory of field and lab equipment, maintained accurate supply records, and placed orders to prevent stockouts
- Developed R scripts for QA/QC of ecological datasets after field work, improving data integrity and workflow efficiency

Fuels Monitoring Technician

Great Basin Institute (GBI) / Bureau of Land Management: Ely, NV — Jun 2017–Oct 2017, full time

- Conducted vegetation and fuels assessments using Assessment, Inventory, and Monitoring (AIM) protocols
- Delivered field data for fire ecology and fuels management programs across BLM-managed lands

EDUCATION

University of Arizona - M.S. Data Science (Online), Expected Dec 2026

University of Arizona - B.S. Natural Resources: Wildlife Conservation & Management, May 2017

PROGRAMMING PROJECTS

NEON Small Mammal Tracker

An R Shiny app turning National Ecological Observatory Network box-trapping records into the life history of every small mammal NEON has caught across 46 field sites. Open a tap-a-site national map to see who lives there, or flip to "by species" to map one animal's range across the country. Drill into any site for a Hall of Fame of the most-caught individuals, trap-grid home-range heatmaps, Hill-number diversity profiles, and detection-corrected closed-capture

abundance estimates (Schnabel/Chapman) that count the animals the traps missed. Includes shareable trading-card profiles, a two-site comparison, and a printable report card. A complete redesign of the original tracker.

[View live app ↗](#)

Big 12 Girth Index - Recruiting Size Analytics

An R Shiny app visualizing every Big 12 football and basketball recruit's height, weight, and star rating from 2016 to present, across all 16 programs. Plot any roster on a Body Map (height × weight, tap a dot to pin a player card), compare position-group size distributions against the conference in Position DNA, project incoming defenders onto 3-3-5 roles in the War Room, and track class quality across coaching eras. Data scraped from 247Sports; season records and SP+ from CollegeFootballData. A full rebuild and major expansion of the earlier Big-12 Talent Pathways app.

[View live app ↗](#)

NEON Ground Beetle Tracker

An R Shiny app exploring ground beetle (Carabidae) biodiversity across all 46 NEON terrestrial sites using pitfall-trap records from NEON DP1.10022.001. Ground beetles are textbook bioindicators (they respond fast to habitat change and climate) so their richness, diversity, and seasonal activity tell a rich story about each site. Tabs cover community composition with natural-history cards, Hill-number diversity and Hurlbert rarefaction, monthly activity-density seasonality, inter-annual catch-per-effort trends with an insect-decline verdict, and a biogeography view with a national richness map, Bray–Curtis PCoA ordination, and Dufrene–Legendre indicator-species table. All 46 sites are precomputed into a fast-loading bundle; effort is normalized to catch per 100 trap-nights throughout. The carabid sibling of the NEON Small Mammal Tracker.

[View live app ↗](#)

NEON Water Chemistry Analyte Viewer

An R Shiny app comparing surface-water chemistry across 34 NEON aquatic field sites from desert streams to Alaskan lakes to Puerto Rican rivers, on roughly 197,000 real measurements. Pick any two analytes and explore a normalized or dual-axis time series, a least-squares relationship fit with $R^2/p/n$ and a temporal-autocorrelation flag, a Spearman correlation screen with per-row sample sizes, a real STL seasonal decomposition, and a small predictor that estimates an analyte from its best-correlated neighbors and scores it against a mean-only baseline. Below-detection values are flagged rather than hidden, and every statistic shows its n. Loads instantly from a pre-bundled dataset.

[View live app ↗](#)

Which State's Flora Matches Each Movie World Best?

An R Shiny app asking which U.S. state's plant life best fits three cinematic biomes (Dune's Arrakis, Lord of the Rings' Middle-earth, and Jurassic Park's Isla Nublar), each defined by a curated set of plant families. Select any states and an interactive bar chart ranks their flora against each world; toggle Raw counts → Fair share to correct for the fact that large families like Asteraceae (~65% of Arizona's Middle-earth score) would otherwise dominate, with hover tooltips showing the exact families driving each score. A chord diagram then maps species shared across states and worlds. Built on ~250,000 USDA PLANTS records precomputed into an 11 KB bundle for instant load.

[View live app ↗](#)

REFERENCES

Donald E. Merson

Data Scientist Manager III

Web Applications Developer IV — VGS project (University of Arizona)

Tucson, AZ

(520) 282-1188 · dmerson@arizona.edu

Charles Perry

Data Scientist Manager III

PI for VGS project (University of Arizona)

Tucson, AZ

(517) 648-6746 · csperry@arizona.edu

Abe Karam

Domain Manager, Domain 14 (Arizona and New Mexico)

National Ecological Observatory Network

Tucson, AZ (Sycamore Creek)

(541) 840-0066 · akaram@battelleecology.org

Rachel More-Hla

Limnology and Aquatic Instrumentation Technician Specialist

National Ecological Observatory Network

Tucson, AZ (Sycamore Creek)

(520) 444-8409 · ram5@arizona.edu